WoSM Academy

MAT231-F27 · CALCULUS II

Change of variables

Polar, cylindrical and spherical coordinates

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Lecture 20 · Unit 20 · Fall 2027

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Last time

- Triple integrals: Riemann sums over boxes; Fubini in any of six orders.
- A solid between two surfaces over its shadow; slicing by planes.
- Mass, centre of mass, moments of inertia, probabilities as volumes.

Discs, balls, cylinders and cones gave square roots in the limits. Today: coordinates in which these shapes become boxes, and the factor that every change of coordinates brings with it.

This lecture

01 Polar coordinates

02 The change of variables formula

03 Cylindrical coordinates

04 Spherical coordinates

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.3, 5.5 and 5.7.

01

Polar coordinates

Discs are awkward in Cartesian coordinates

The volume under $z = 4 - x^2 - y^2$ over the disc $x^2 + y^2 \le 4$ (Lecture 19):

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - x^2 - y^2)\, dy\, dx.$$

Possible, but unpleasant.

In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ the disc is the rectangle

$$0 \le r \le 2, \qquad 0 \le \theta \le 2\pi,$$

and the integrand is $4 - r^2$. What is dA in terms of dr and $d\theta$?

The area of a polar rectangle

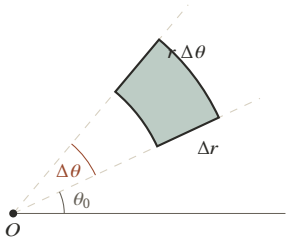
The region $r_0 \le r \le r_0 + \Delta r$, $\theta_0 \le \theta \le \theta_0 + \Delta\theta$ is a piece of an annulus. Its area is

$$\tfrac{1}{2}\big((r_0 + \Delta r)^2 - r_0^2\big)\,\Delta\theta = \bar{r}\,\Delta r\,\Delta\theta,$$

where $\bar{r} = r_0 + \frac{\Delta r}{2}$ is the middle radius. It is almost a rectangle with sides Δr and $r\,\Delta\theta$. So

$$dA = r\,dr\,d\theta.$$

Far from the origin, the same $\Delta\theta$ sweeps out more area: hence the factor r.



Double integrals in polar coordinates

Theorem

If $D = \{(r\cos\theta, r\sin\theta) : \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$ with $0 \le h_1 \le h_2$ and $\beta - \alpha \le 2\pi$, and f is continuous on D, then

$$\iint_D f\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta.$$

- Replace x by $r\cos\theta$ and y by $r\sin\theta$ — and dA by $r\, dr\, d\theta$. Forgetting the r is the most common mistake.
- The proof: Riemann sums over polar rectangles, whose areas are $\bar{r}\,\Delta r\,\Delta\theta$.
- A general form for any change of coordinates follows shortly.

Examples

Example (The volume from before)

$$\int_0^{2\pi}\int_0^{2}(4-r^2)\,r\,dr\,d\theta = 2\pi\left[2r^2-\frac{r^4}{4}\right]_0^2 = 2\pi\,(8-4) = 8\pi,$$

as found by slicing in Lecture 19.

Example (The annulus $1 \le x^2+y^2 \le 4$)

$$\iint (x^2+y^2)\,dA = \int_0^{2\pi}\int_1^{2} r^2\cdot r\,dr\,d\theta = 2\pi\cdot\frac{16-1}{4} = \frac{15\pi}{2}.$$

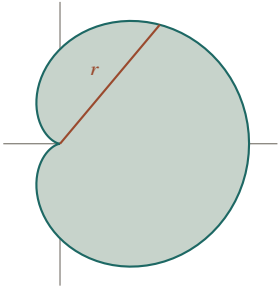
Example: the cardioid

The region inside $r = 1 + \cos\theta$, $0 \le \theta \le 2\pi$:

$$\text{area} = \int_0^{2\pi} \int_0^{1+\cos\theta} r \, dr \, d\theta = \frac{1}{2} \int_0^{2\pi} (1 + \cos\theta)^2 \, d\theta.$$

Expand: $\int_0^{2\pi} (1 + 2\cos\theta + \cos^2\theta) \, d\theta = 2\pi + 0 + \pi$. So the area is $\frac{3\pi}{2}$.

The formula $\frac{1}{2} \int r(\theta)^2 \, d\theta$ for the area of a region bounded by a polar curve is the special case $f = 1$, $h_1 = 0$.



$r = 1 + \cos\theta$

02

The change of variables formula

Substitution in one variable

For a C^1 function g that is increasing or decreasing on $[a, b]$:

$$\int_{g([a,b])} f(x)\,dx = \int_a^b f\big(g(u)\big)\,|g'(u)|\,du \qquad \text{(Calculus I, Unit 21).}$$

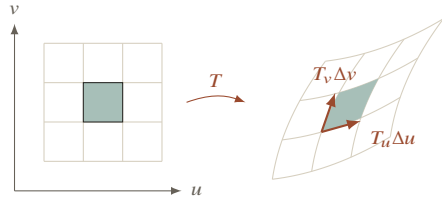
- $|g'(u)|$ is the **stretch factor**: g maps a short interval of length Δu to one of length about $|g'(u)|\,\Delta u$.
- In two variables a map $T(u, v) = \big(x(u, v), y(u, v)\big)$ sends small rectangles to small (curved) parallelograms. The factor is the ratio of their areas.

The Jacobian as an area factor

T maps the small rectangle $[u, u+\Delta u] \times [v, v+\Delta v]$ to a region that is almost the parallelogram spanned by $T_u\,\Delta u$ and $T_v\,\Delta v$ (since $T(u+\Delta u, v) - T(u,v) \approx T_u\,\Delta u$). Its area is $|\det(T_u, T_v)|\,\Delta u\,\Delta v$ (Lecture 12). With the **Jacobian**

$$\frac{\partial(x,y)}{\partial(u,v)} = \det DT = \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix},$$

$$dA = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$



The change of variables theorem

Theorem

Let $T\colon S \to D$ be a C^1 map from a region S in the uv-plane onto a region D in the xy-plane, one-to-one with $\det DT \neq 0$ except possibly on the boundary of S. Then for continuous f

$$\iint_D f(x,y)\,dA = \iint_S f\big(T(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

- Polar: $\frac{\partial(x,y)}{\partial(r,\theta)} = r$ (Lecture 17), which is 0 only on the boundary $r = 0$.
- The same holds in $\mathbb{R}^n$, with $|\det DT|$ the volume factor.
- Choose T so that S is simple (a rectangle), or the integrand is.

Example: a linear change

D is the square bounded by the lines $x + y = 0$, $x + y = 2$, $x - y = 0$, $x - y = 2$. Compute $\iint_D (x^2 - y^2)\,dA$.

- Put $u = x + y$, $v = x - y$. Then $S = [0, 2] \times [0, 2]$, and $x^2 - y^2 = uv$.
- $x = \frac{u+v}{2}$, $y = \frac{u-v}{2}$: $\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} = -\frac{1}{2}$.

$$\iint_D (x^2 - y^2)\,dA = \int_0^2 \int_0^2 uv \cdot \frac{1}{2}\,du\,dv = \frac{1}{2} \cdot 2 \cdot 2 = 2.$$

Example (The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1$)

$x = au$, $y = bv$ maps the unit disc onto it, with Jacobian ab. So its area is $ab \cdot \pi$.

Using the inverse Jacobian

Often u, v are given as functions of x, y. By the inverse function theorem (Lecture 17), $\frac{\partial(x,y)}{\partial(u,v)} = 1 \big/ \frac{\partial(u,v)}{\partial(x,y)}$.

Example. The region D in the first quadrant between $xy = 1$, $xy = 3$, $y = x$, $y = 2x$.

- Put $u = xy \in [1, 3]$ and $v = y/x \in [1, 2]$: D becomes a rectangle.
- $\frac{\partial(u,v)}{\partial(x,y)} = \det \begin{pmatrix} y & x \\ -y/x^2 & 1/x \end{pmatrix} = \frac{2y}{x} = 2v$, so $dA = \dfrac{du\,dv}{2v}$.
- $\text{area}(D) = \displaystyle\int_1^3 \int_1^2 \frac{dv}{2v}\,du = 2 \cdot \frac{\ln 2}{2} = \ln 2.$

No need to solve for x and y (which gives $x = \sqrt{u/v}$, $y = \sqrt{uv}$).

03

Cylindrical coordinates

Cylindrical coordinates

Polar coordinates in the xy-plane, and z unchanged:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z; \qquad \frac{\partial(x,y,z)}{\partial(r,\theta,z)} = r, \quad dV = r\,dr\,d\theta\,dz.$$

Good for solids with an axis of symmetry: cylinders, cones, paraboloids.

Example (The paraboloid solid, a third time)

$r^2 \le z \le 4$ over $0 \le r \le 2$:

$$\int_0^{2\pi}\int_0^2\int_{r^2}^4 r\,dz\,dr\,d\theta = 2\pi\int_0^2 (4r - r^3)\,dr = 8\pi.$$

Examples in cylindrical coordinates

Example (The cone $\sqrt{x^2+y^2} \le z \le 1$, that is $r \le z \le 1$)

$$\int_0^{2\pi}\int_0^1\int_r^1 r\,dz\,dr\,d\theta = 2\pi\int_0^1 (r-r^2)\,dr = \frac{\pi}{3} = \frac{1}{3}\cdot \text{base}\cdot\text{height}.$$

Example (A solid cylinder: radius a, height h, density 1)

Its moment of inertia about its axis ($M = \pi a^2 h$):

$$I = \int_0^{2\pi}\int_0^a\int_0^h r^2\cdot r\,dz\,dr\,d\theta = \frac{\pi a^4 h}{2} = \frac{1}{2}Ma^2.$$

04

Spherical coordinates

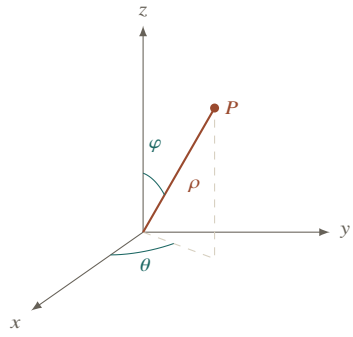
Spherical coordinates

$$x = \rho \sin\varphi \cos\theta, \qquad y = \rho \sin\varphi \sin\theta, \qquad z = \rho \cos\varphi$$

- $\rho \geq 0$: the distance to O;
- $0 \leq \varphi \leq \pi$: the angle from the z-axis;
- $0 \leq \theta \leq 2\pi$: as in polar coordinates.

The Jacobian determinant is $\rho^2 \sin\varphi \geq 0$ (a small spherical box has edges $\Delta\rho$, $\rho\,\Delta\varphi$, $\rho \sin\varphi\,\Delta\theta$):

$$dV = \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta.$$



Examples in spherical coordinates

Example (The ball of radius R)

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{R} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta = 2\pi \cdot 2 \cdot \frac{R^3}{3} = \frac{4}{3}\pi R^3.$$

Example ($\iiint e^{(x^2+y^2+z^2)^{3/2}}\, dV$ over the unit ball)

$$2\pi \cdot 2 \cdot \int_0^1 e^{\rho^3} \rho^2 \, d\rho = 4\pi \cdot \frac{e-1}{3}$$

— hopeless in Cartesian coordinates.

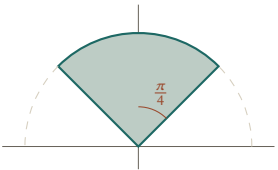
Example: the ice-cream cone

The solid inside the sphere $\rho \le 1$ and within the cone $\varphi \le \frac{\pi}{4}$, that is $z \ge \sqrt{x^2 + y^2}$.

$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$

$$= 2\pi\left(1 - \frac{\sqrt{2}}{2}\right)\frac{1}{3} = \frac{\pi\,(2 - \sqrt{2})}{3} \approx 0.613.$$

In spherical coordinates the region is a box: $[0, 1] \times [0, \frac{\pi}{4}] \times [0, 2\pi]$.



A cross-section through the z-axis.

Example: the centroid of a hemisphere

The solid hemisphere $\rho \le R$, $0 \le \varphi \le \frac{\pi}{2}$ has volume $\frac{2}{3}\pi R^3$, and $\bar{x} = \bar{y} = 0$ by symmetry.

$$\iiint z\,dV = \int_0^{2\pi}\int_0^{\pi/2}\int_0^{R} (\rho\cos\varphi)\,\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = 2\pi\cdot\frac{1}{2}\cdot\frac{R^4}{4} = \frac{\pi R^4}{4}.$$

$$\bar{z} = \frac{\pi R^4/4}{2\pi R^3/3} = \frac{3R}{8}.$$

The centroid is below the middle height $\frac{R}{2}$: most of the volume is near the flat base.

Which coordinates?

Region	Coordinates
discs, annuli, sectors, polar curves	polar
cylinders, cones and paraboloids around an axis	cylindrical
balls, spherical shells, cones with vertex at O	spherical
parallelograms, regions between level curves	a linear or adapted change

- Look at the region first, then at the integrand: $x^2 + y^2$ suggests polar or cylindrical, $x^2 + y^2 + z^2$ spherical.
- Always include the Jacobian factor: r, r, $\rho^2 \sin\varphi$, or $|\det DT|$.

05

Summary and practice

Summary

- **Change of variables:** $\iint_D f\,dA = \iint_S f(T(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$; the Jacobian determinant is the area (volume) factor.
- Polar: $dA = r\,dr\,d\theta$. Cylindrical: $dV = r\,dr\,d\theta\,dz$. Spherical: $dV = \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta$.
- Choose coordinates in which the region is a rectangle or box, or the integrand simplifies.
- If u, v are given in terms of x, y, use $\frac{\partial(x,y)}{\partial(u,v)} = 1\Big/\frac{\partial(u,v)}{\partial(x,y)}$.

Practice

1. Compute $\iint \sqrt{x^2 + y^2}\, dA$ over the disc $x^2 + y^2 \leq 9$.
2. Find the area inside the circle $r = 2 \sin \theta$.
3. Compute $\iint_D (x + y)^2\, dA$, where D is the square $|x| + |y| \leq 1$.
4. Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$.
5. Find the volume inside the cylinder $x^2 + y^2 \leq 1$, above $z = 0$ and below $z = 4 - x^2 - y^2$.
6. Find the mass of the unit ball if the density is the distance to the centre.
7. Find the volume between the spheres of radius 1 and 2 (centred at O) within the cone $\varphi \leq \frac{\pi}{3}$.

Answers

1. $\int_0^{2\pi}\int_0^{3} r\cdot r\,dr\,d\theta = 2\pi\cdot 9 = 18\pi$.
2. $\frac{1}{2}\int_0^{\pi} 4\sin^2\theta\,d\theta = \pi$ (a circle of radius 1).
3. $u = x+y$, $v = x-y$ range over $[-1,1]^2$, with $|J| = \frac{1}{2}$: $\frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} u^2\,du\,dv = \frac{2}{3}$.
4. $x = au$, $y = bv$, $z = cw$ maps the unit ball onto it, with Jacobian abc: $\frac{4}{3}\pi abc$.
5. $\int_0^{2\pi}\int_0^{1}(4-r^2)\,r\,dr\,d\theta = 2\pi\left(2-\frac{1}{4}\right) = \frac{7\pi}{2}$.
6. $\int_0^{2\pi}\int_0^{\pi}\int_0^{1}\rho\cdot\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = 4\pi\cdot\frac{1}{4} = \pi$.
7. $2\pi\left(1-\cos\frac{\pi}{3}\right)\cdot\frac{2^3-1}{3} = \frac{7\pi}{3}$.

Next time

Lecture 21 — The Gaussian integral; improper multiple integrals (Unit 21)

- $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$, by a double integral in polar coordinates
- When do integrals over unbounded regions, or of unbounded functions, converge?
- The normal distribution, and the volume of a ball in n dimensions

Reading: Marsden & Tromba, *Vector Calculus*, Section 6.4.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.